

MIT Art Design and Technology University's
MIT School of Computing, Pune
Department of Computer Science and Engineering
BTech Third Year A.Y.2025-26



Machine Learning Lab

Enrolment No: _____

Division: _____

Roll No: _____

Name: _____

Experiment No.: 08

Title: To study working of Logistic Regression and visualize the results.

Theory:

Logistic Regression is a supervised machine learning algorithm used for classification problems. Unlike Linear Regression which predicts continuous values, Logistic Regression predicts the probability of a data point belonging to a certain class.

It works by applying the sigmoid (logistic) function on a linear combination of input features:

$$P(y = 1|x) = \frac{1}{1 + e^{-(w^T x + b)}}$$

The output lies between 0 and 1, representing probability.

A threshold (usually 0.5) is applied to classify the data into two categories.

For multiclass problems, Logistic Regression uses the One-vs-Rest (OvR) or Softmax approach.

Advantages:

Simple and easy to implement.

Provides probabilistic interpretation.

Works well for linearly separable data.

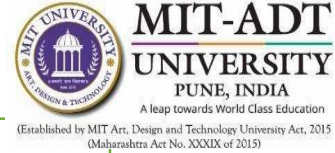
Applications:

Medical diagnosis (disease vs no disease).

Spam email detection.

Iris dataset flower classification.

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	In this experiment: - The Logistic Regression model is trained on a dataset (e.g., Iris). - Predictions are made and results are visualized using: Sigmoid curve (probability vs feature). Decision boundary plots (for 2D features). Confusion matrix to evaluate accuracy.
Pseudo code or Program	<pre> import numpy as np import matplotlib.pyplot as plt from sklearn.datasets import load_iris from sklearn.linear_model import LogisticRegression from sklearn.model_selection import train_test_split from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay, classification_report # Load dataset iris = load_iris() X = iris.data[:, 2].reshape(-1, 1) # Petal length (1D feature) y = (iris.target == 0).astype(int) # Binary classification: Setosa (1) vs others (0) # Train/test split X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42) # Train logistic regression clf = LogisticRegression() clf.fit(X, y) y_pred = clf.predict(X_test) # --- Confusion Matrix --- cm = confusion_matrix(y_test, y_pred) disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["Not Setosa", "Setosa"]) disp.plot(cmap="Blues") plt.title("Confusion Matrix") plt.show() </pre>

	<pre> # --- Classification Report --- print("Classification Report:\n", classification_report(y_test, y_pred)) # Create range of values for smooth curve X_test = np.linspace(X.min(), X.max(), 300).reshape(-1, 1) y_prob = clf.predict_proba(X_test)[:, 1] # Probability of being Setosa # Plot sigmoid curve plt.scatter(X, y, c=y, cmap=plt.cm.coolwarm, edgecolor="k", s=50, label="Data") plt.plot(X_test, y_prob, color="black", linewidth=2, label="Sigmoid curve") plt.xlabel("Petal length (cm)") plt.ylabel("Probability of Setosa") plt.title("Logistic Regression with Sigmoid Curve (Iris)") plt.legend() plt.show()plt.ylabel("Petal Length (cm)") plt.title("Linear Regression on Iris Dataset") plt.legend() plt.show() </pre>
Outputs	(Students can attach extra pages if necessary)

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Exercise for Practice	
Q1.	What is Logistic Regression used for?
Ans:	
Q2.	How does Logistic Regression differ from Linear Regression?
Ans	
Q3	What is the role of the sigmoid function in Logistic Regression?
Ans	
Q4.	Can Logistic Regression handle multiclass classification?
Ans	

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Q5	What are the advantages of Logistic Regression?
Ans	
Conclusion	The experiment demonstrated the working of Logistic Regression for classification. Using the sigmoid function, the model successfully mapped inputs to probabilities and classified data. Visualization and confusion matrix confirmed that Logistic Regression is effective for simple classification tasks.

Date of Submission:	
Marks out of 10	Name and Sign of Subject Teacher
Remark:	

